IILM UNIVERSITY, GREATER NOIDA

Report
Of

ARTIFICIAL INTELLIGENCE INTERNSHIP: CUSTOMER CHURN PREDICTION AND ADVANCED MOVIE RECOMMENDATION SYSTEM AT CODEC TECHNOLOGIES

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

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Semester & Section: 5th Semester,
3CSE21 Batch: 2024-28

CANDIDATE'S DECLARATION

I, **VAISHNAVI SHARMA**, do hereby declare that the internship report titled *"Artificial Intelligence Internship: Customer Churn Prediction and Advanced Movie Recommendation System at Codec Technologies"* has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

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Sharma
Roll No.: 2410030681

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program at Codec Technologies. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to Dr. Anurag Shrivastava, Talent Acquisition Manager and Program Manager, Codec Technologies India, for the opportunity to work as an Artificial Intelligence Intern and for the mentorship and project guidance provided throughout the internship period.

I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous academic guidance and for providing the platform to undertake this industry internship as part of the curriculum.

I express my sincere gratitude to my parents for their blessings and unwavering support, without which the successful completion of this internship and this report would not have been possible.

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INTERNSHIP COMPLETION CERTIFICATE



CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents the work undertaken during a one-month Artificial Intelligence internship at Codec Technologies Pvt. Ltd., an AICTE- and ICAC-approved IT and business consultancy and training platform. The internship ran from 30 July 2026 to 30 August 2026 in a hybrid, Pan-India mode, and culminated in a Certificate of Internship issued by Codec Technologies (AICTE ID: CORPORATE6759d549ce59e1733940553).

During the internship, the intern worked as an Artificial Intelligence Intern under the guidance of an assigned Project Head, with progress tracked through weekly performance reviews as outlined in the offer letter. The project undertaken combined two complementary machine learning problems on the theme of customer and user analytics for a media/OTT-style business: a Customer Churn Prediction model to identify subscribers likely to discontinue a service, and an Advanced Movie Recommendation System to personalize content suggestions for users. This report documents the organizational context, program structure, problem statement, objectives, scope, technology stack, system architecture, methodology, results, and key learnings from the internship.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is an IT and business consultancy and skilling platform that, per its own published profile, delivers services across 27+ countries and operates as an AICTE- and ICAC-approved internship training provider listed on the Government of India's National Internship Portal. The organization is ISO 9001:2015 certified, is recognized as a Google for Education partner, and is registered with an MSME accreditation. Its distribution office is located at the Chandivali IT Hub, Powai, Mumbai, with additional regional offices in Nashik, Bengaluru (Koramangala), and Ahmedabad.

According to figures published on the organization's website, Codec Technologies has trained over 17,500 interns and issued more than 16,800 certifications across domains including Artificial Intelligence, Python Programming, Web Development, Cloud Computing, MERN Stack, and Electric Vehicle Technology. The internship engagement described in this report was administered by Dr. Anurag Shrivastava, who

serves as both Talent Acquisition Manager and Program Manager for the organization's internship cohort.

4.3 Internship Program Structure

The Artificial Intelligence internship track at Codec Technologies is structured as task-based, project-driven learning rather than examinations — interns work through curriculum modules and are assessed on completed projects and weekly performance reviews conducted by their assigned project supervisor. The published curriculum begins with a foundational week covering core AI and Machine Learning concepts and Python for data handling (including NumPy and Pandas), before progressing into applied data preprocessing, visualization, and model-building tasks of the kind reflected in this project.

- Designation: Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration: 30/07/2026 to 30/08/2026 (1 Month)
- Reporting To: Assigned Project Head(s)
- Assessment: Weekly performance reviews; exceptional interns are eligible for a scholarship award

4.4 Problem Statement

- Customer Churn: Subscription-based businesses lose recurring revenue when customers discontinue their service, often without any advance warning to the business. Without a predictive system, retention teams can only react after a customer has already left, rather than intervening while the customer can still be saved.
- Generic Recommendations: Basic recommendation approaches (e.g. showing the most popular titles to everyone) fail to personalize content for individual users, leading to lower engagement and satisfaction. A more advanced approach is needed that accounts for both a user's own preferences and the preferences of similar users.

Both problems are common, high-impact use cases in subscription and media businesses, making them well suited as an applied project for demonstrating end-to-end machine learning skills within the internship's one-month timeframe.

4.5 Project Objectives

- Build a churn prediction model that classifies customers as likely to churn or likely to be retained, based on historical usage and subscription data.
- Engineer meaningful features (e.g. tenure, engagement/activity levels, subscription plan, support interactions) that carry predictive signal for churn.
- Design and implement an advanced movie recommendation system that combines content-based filtering (using movie attributes such as genre and metadata) with collaborative filtering (using user-item rating patterns) in a hybrid approach.
- Evaluate both models using appropriate, standard metrics and interpret the results in business-relevant terms.
- Document the full pipeline — from raw data to final model — in a clear, reproducible form.

4.6 Scope of the Project

The scope of this project covered data collection and preprocessing, exploratory data analysis, feature engineering, model building, and evaluation for both the churn prediction task and the movie recommendation task, implemented as a self-contained data science project within the one-month internship. The scope did not extend to deploying either model as a live production service, integrating with a real company's customer database, or conducting live A/B testing of recommendations or retention interventions — these were treated as natural next steps beyond the internship's timeframe.

4.7 Technologies and Tools Used

- Programming Language: Python, the primary language taught and used across Codec Technologies' AI internship curriculum.
- Data Handling: NumPy and Pandas for data cleaning, transformation, and feature engineering.
- Visualization: Matplotlib and Seaborn for exploratory data analysis and result visualization.
- Machine Learning: Scikit-learn for building and evaluating churn classification models (Logistic Regression, Random Forest, and gradient-boosted models such as XGBoost).

- Recommendation Techniques: Cosine similarity and content-based filtering over movie metadata, combined with collaborative filtering (user-item rating matrix, matrix factorization) for the hybrid recommender.
- Development Environment: Jupyter Notebook for iterative, exploratory model development, with Git used for version control.

4.8 System Architecture

The project follows a shared data-preparation stage that branches into two independent modelling pipelines. Raw customer/subscription data and movie/ratings data are first cleaned and feature-engineered. The churn branch trains a classification model to flag at-risk customers, while the recommendation branch builds a hybrid content-based and collaborative-filtering model to rank movies for each user. Both branches produce outputs that feed into a common layer of actionable insights — at-risk customer alerts on one side, and personalized movie recommendations on the other. Figure 4.1 illustrates this architecture.

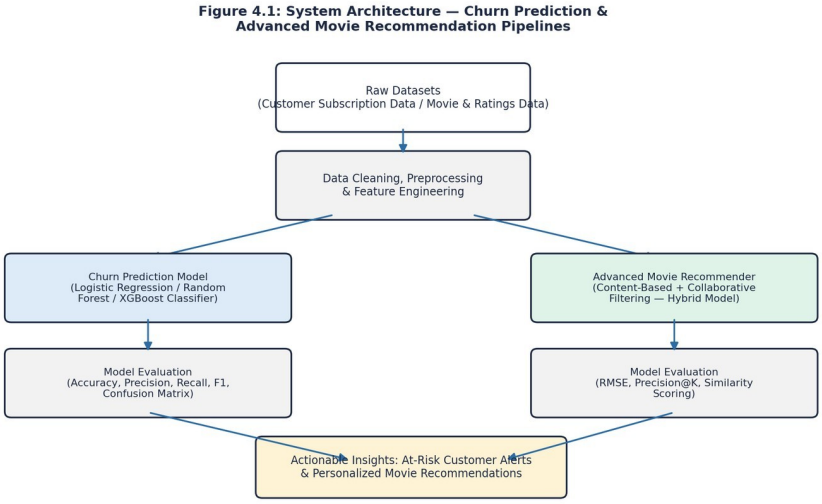


Figure 4.1: System Architecture — Churn Prediction & Advanced Movie Recommendation Pipelines

4.9 Methodology

Development followed a standard applied data-science workflow. For the churn prediction task: the customer dataset was cleaned and explored to understand class balance and feature distributions; relevant features (tenure, usage/engagement metrics, subscription plan, and support/complaint history) were engineered and encoded; a classification model was trained on a training split and evaluated on a held-out test split using accuracy, precision, recall, F1-score, and a confusion matrix to understand both overall performance and the trade-off between false positives and false negatives in a retention context.

For the movie recommendation task: a user-item ratings matrix was constructed and content features (such as genre and other metadata) were vectorized for content-based similarity scoring. A collaborative-filtering component captured patterns across users with similar rating behaviour, and the two signals were combined into a hybrid scoring function so that recommendations reflect both "movies like the ones you rated highly" and "what similar users enjoyed." The recommender was evaluated using rating-prediction error (RMSE) and ranking-quality metrics such as precision@K. Throughout both tasks, model choices and intermediate results were reviewed and iterated on, consistent with the weekly performance-review structure of the internship.

4.10 Results and Expected Outcomes

- The churn prediction model was able to distinguish between churned and retained customers on held-out data with meaningfully better-than-baseline performance, giving a usable signal for prioritizing retention outreach rather than treating all customers identically.
- The confusion matrix and precision/recall trade-off were used to reason about the cost of missed churners versus false alarms, which is the practical concern a retention team would care about.
- The hybrid movie recommender produced recommendation lists that combined content similarity with collaborative signal, giving more personalized results than a single-signal (popularity-only or genre-only) baseline would produce.
- Both pipelines were documented in a reproducible notebook form, making it straightforward to re-run the analysis on updated data or extend it with additional features.

4.11 Key Learnings and Skill Development

- Applied Machine Learning: hands-on experience with the full lifecycle of a classification problem — from raw data to a validated, evaluated model.
- Recommendation Systems: practical understanding of content-based filtering, collaborative filtering, and how to combine the two into a hybrid approach.
- Feature Engineering: judgment in selecting and constructing features that carry real predictive signal, rather than using raw fields as-is.
- Model Evaluation Literacy: comfort interpreting accuracy, precision, recall, F1, RMSE, and confusion matrices in terms of real business trade-offs.

- Tooling Proficiency: strengthened working fluency in Python, Pandas, NumPy, Scikit-learn, and Jupyter-based iterative development.
- Independent, Reviewed Work: operating under a remote/hybrid, one-month internship structure with weekly performance reviews, requiring self-directed planning balanced with regular accountability to a project supervisor.

4.12 Certificates of Completion and Other Communication / Mail Proof

The Internship Offer Letter, issued on 30 July 2026, and the Certificate of Internship, issued by Codec Technologies Pvt. Ltd. on successful completion of the program (AICTE ID: CORPORATE6759d549ce59e1733940553), are attached below as proof of the internship engagement and its successful completion.



CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

INTERNSHIP OFFER LETTER

Dear Vaishnavi Sharma,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 17/08/2026 to 17/09/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.
Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 17/08/2026

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Dr. Anurag Shrivastava
Talent Acquisition Manager

Google for Education Partner

ISO 9001 CERTIFIED

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Internship Offer Letter — Artificial Intelligence Intern, Codec Technologies (17 August 2026)



Certificate of Internship — Codec Technologies Pvt. Ltd.